



Yasuda Koichiro <k.yasuda5160@gmail.com>

about your boldface question on page

40 件のメッセージ

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月18日 7:00

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

Danny: { \bf Introduces bias/dispersion, but the truncation was to reduce errors in the calculation of the flux because the nice step structure was failing unless we truncated? Should we explain why we truncate some of the response functions? }

\GG{GG: Which errors you said we tried to avoid? We already explained why we cut off the $\delta \Phi$ at 2 MeV - the contribution of neutrinos above is δ_i and our signal is below. Cutting arbitrarily the CurlyR introduced errors since in our procedure the complete kernels CurlyH are the components of the generating vectors of our convex hall, truncating these introduces errors in our fundamental arguments }

Graciela

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月18日 7:06

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

I think when we did not truncate the kernels, the step structure was unstable.

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月18日 10:33

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

See below.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月18日 10:38

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

I cannot explain why we had unstable results. There were numerical issues so frequently that it may have been a numeric error instead of conceptual one. But it could also have been a conceptual error: we were double counting the ϕ above 2 MeV both as signal and as background???

However, I was referring to the error in our convex hall theorems: in cutting some kernels at 2 MeV for sure we are making an error, since we are changing some of the kernels, which are the components of the generating vectors of the convex hall of rates, we are changing the conditions of the theorems we used. I do not know how this would translate into a bad fit.

Graciela

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月18日 10:40

To: Danny Marfatia <dmarf8@hawaii.edu>
Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

Was unstable because of a numerical error or because we were double counting?

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>
To: Graciela Gelmini <gelmini@physics.ucla.edu>
Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

2026年9月18日 10:44

Recall that the fits were unstable and we traced that to the fact that the kernels had unusual shapes above 2 MeV. We truncated the kernels to stabilize the fits, i.e, we were trying to *reduce the error*. We can resubmit once we converge on an accurate statement.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>
To: Danny Marfatia <dmarf8@hawaii.edu>
Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

2026年9月18日 10:54

By the way, this explanation of why we cut at 2 MeV, should go to where we say we fit only the Φ up to 2 MeV, since $j_{\max}$ is only = 2 MeV and we use only the $R_{(i, j)}$ in Figure 5, where you can clearly see some of the R_i are cutoff.

So the explanation should go there.

About the fit being poor near the $j_{\max}$, I think it is because we cutoff some of the R_i (E_{ν}).

Graciela

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>
To: Danny Marfatia <dmarf8@hawaii.edu>
Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

2026年9月18日 10:57

I do not understand what you want to say. But please include the explanation to where it goes when we introduced $\Delta\Phi$ and said we are going to fit only the flux up to 2MeV and treat the h_i as background.

Were we been double counting before?

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>
To: Danny Marfatia <dmarf8@hawaii.edu>
Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

2026年9月18日 11:07

The additional reason you want to give to cut our fit at 2 MeV (which I thought we had already explained) should go when we do the cut, on page 12. This is when we introduce $\Delta\Phi$ (and h_i) as background. having h_i as background due to a known neutrino flux at higher energy is incompatible with fitting Φ at high energy which means treating it as unknown.

What this cut does to the fit we made close to the cut, namely that cutting off some kernels leads to an error (in the convex hull definition we are using) is what should go to explain the bad quality of the fit.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>
To: Danny Marfatia <dmarf8@hawaii.edu>
Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

2026年9月18日 11:12

Another change you made was to a sentence which now I put back as "A different choice of data bins with a larger number d of them, would likely result in a best-fit function with a larger number of downward steps (since their maximum number is $d-1$), and consequently shorter constant segments, rendering the bias closer to zero at the end of each of these segments." You had "choice of data bins, say with a larger number" as meaning for example with larger number of data bins, but I think it is ONLY with a larger number of data bins that there could be more downward steps and thus shorter constant pieces to decrease the bias.

Best,
Graciela

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月18日 11:15

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

I only tweaked your sentence ... did not change its meaning. The "say, with a larger number" was yours.

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月18日 11:17

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

... and actually your original statement was correct, because there's more than one way of modifying the bins, including a different nonuniform binning.

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月18日 11:22

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

Move the text where you want it, but we still have to agree on its content.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月18日 11:48

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

Sorry, I may have made a mistake. It happens.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月18日 11:49

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

Uniform or non-uniform does not matter for the number of downward steps, only its number. With larger d the shorter one expects to be the constant pieces.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月18日 11:52

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

I do not think we should add any sentence: we already explained why we cut our fit at 2 MeV earlier:
we separated known from unknowns and treated the known neutrino spectrum as producing a background.

When talking about why the fit is not good, we mention the effect the cut has not why we made the cut.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月18日 12:02

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

Danny, please do not keep changing the sentence which should be "A different choice of data bins with a larger number of them" that the bins are irregular or regular does not count for the number of downward steps. I wanted to say that increasing the constant pieces will likely be shorter.

I do not think we need to add or modify anything. I did not see any sentence that you added. If you did, could you tell me where?

Graciela

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月18日 12:05

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

The "say" indicates an example of a different binning, in this case more bins. There are other possibilities that may reduce the bias without increasing the number of bins. You can change it back.

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月18日 12:08

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

I changed it back. After you move the text where you want it, I will edit it and we can see if we agree.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月18日 12:16

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

If you add a sentence at this point, please mark it with bf so I can see it. As I said:

earlier we explained why we cut our fit at 2 MeV (after we treat the spectrum as known and the h_i as background) do not need to add anything else.

later, the cut of some R_i ($e\nu$ at 2 MeV) may have introduced an error that translated into a bad fit. Not sure but it is likely.

All is said. We do not need to add anything.

Unless you add/change a sentence somewhere, I think the paper is ready for submission

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月18日 12:18

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

If you want to talk about other possibilities of binning and their effect is leading to shorter constant pieces, please make another sentence. I do not know why other possibilities would lead to a larger number of downward steps, except the number of bins.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月18日 12:20

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

Danny: Explain to me why a different, irregular size of bins, without changing its number will likely lead to more downward steps?

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月18日 12:21

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

I DO NOT WANT TO MOVE THE TEXT ANYWHERE. which text are you talking about?

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月18日 12:23

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

A different binning will not lead to a larger number of downward steps, but could lead to a reduced bias!

Didn't you want to move some text to a different location? Read your emails in this chain. Maybe that's not what you mean to say.

We have not said anywhere that we cut the kernels at 2 MeV to improve our fit.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月18日 12:55

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

Danny: I am only arguing about the length of the segments: if they are shorter, since the bias is zero at the middle it will deviate less. I am ONLY talking about this way in which the bias would be smaller. I do not know about other ways. They are possible but I am not talking about any other way but the length of the constant pieces.

When we cut our fit at 2 MeV and introduced $\Delta\Phi$, I mentioned in our discussions that I did not know what cutting off pieces of the kernels would do to the convex hull theorems at the basis of the method. We are modifying some of the kernels, which are the components of the generator vectors of the convex hull of rates. I do not know what this does, but it is likely that it introduces an error in the part of the hull affected by these cuts.

If you agree how the paper is now please upload it to the arXiv.

Graciela

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月19日 0:20

To: Graciela Gelmini <gelmini@physics.ucla.edu>
Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

Ko, Muping, can you confirm that without truncating the kernels, the step-structure of the best-fit flux was unstable?
[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu> 2026年9月19日 4:31
To: Danny Marfatia <dmarf8@hawaii.edu>
Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

Danny:

Any error was numerical on our part, not due to the method itself. We had a numerical problem when some curlyR values were much larger than the others -- the program could not find steps at high energy. BUT, we had THE SAME PROBLEM at all energies before

Early one we could not find steps (at all) due numerical issues solved in obscure ways: changing from Mathematica routines to Python, multiplying by humongous numbers, changing units... without our full understanding of what they were. I think the problem was one of those.

So to solve our inability of finding the correct numerical method we cutoff having some Curly R that were much larger than the others.

I am mentioning a problem of the method (cutting off the curlyR_i). In the method what type of data points one has does not change the dirac Sum, just its number.

You want to mention in the paper that we cutoff the energy because our program could not deal with different sizes of CurlyR numbers and we were unable to find a better numerical solution?

Graciela
[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu> 2026年9月19日 5:33
To: Graciela Gelmini <gelmini@physics.ucla.edu>
Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

I want Ko/Muping to confirm our reason for truncating the kernels, and then clarify the statement at the bottom of page 19.
[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu> 2026年9月19日 6:44
To: Danny Marfatia <dmarf8@hawaii.edu>
Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

The paper already justifies our choice to fit up to 2MeV in Sec 5. You want to say now that our reason to do so was NOT the one we gave in Sec 5, but that we did it because otherwise we could not make our numerical calculation work.
[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu> 2026年9月19日 6:45
To: Graciela Gelmini <gelmini@physics.ucla.edu>
Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

If that's true, I want to say it ... we were trying to *reduce errors*.

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月19日 8:03

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

If you agree with the boldface on page 19 we are done.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月19日 9:30

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

I do not want to say anything about reducing errors we did not understand, did not show in the paper, did not quantify, may have come from errors in programming and are not related to the method itself. We explained our procedure clearly in sec 5: the maximum value of j , $j = N_{\text{int}}$, corresponds to 2MeV, therefore we do not consider R_{iJ} where j is any higher than 2 MeV. We said it- we showed it in our figure for the $R_{\{iJ\}}$.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月19日 9:33

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

This motivation is NOT what we said or justified in Sec 5- change Sec 5 if you can do better.

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月19日 9:47

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

I did not say anything about reducing errors. Did you read the text?

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月19日 9:55

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

Added a sentence to Section 5 to the same effect.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月19日 11:20

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

NO- Mentioning "stabilizing the step structure" without explaining why it was not stable does not make any sense. Not stabilizing is a numerical error in the minimization procedure- Not of the method itself

Moreover, your extra explanation does not add anything about why the fit we made is not good at high energy.

If you want to change the physical motivation we made in Section 5, this is where you need to explain clearly why any other procedure did not lead to "stabilization" whatever this means, namely why it necessarily led to errors in our

minimization procedure.

Ask Ko and Muping to understand and show what error of "not-stabilization" their program made-
Otherwise, Sec 5 is all the explanation we can give and we SHOULD give.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月19日 11:22

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

NO- an error in our programming should not be our motivation. Sec 5 gives a physical motivation.
Otherwise let us go back and do the work to correct the numerical error and do all the paper again...

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月19日 11:23

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

NO

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>

2026年9月19日 11:25

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

I am not willing to conceal what we did.

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月19日 11:38

To: Danny Marfatia <dmarf8@hawaii.edu>

Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

It is not conceal anything!!!! We had many numerical problems and we do not mention them because they are not relevant to our method. . Do you want to go through all of them? We did not take the time to solve the one we had last and opted for a solution that makes sense physically.

[元のメッセージ非表示]